

HW 2 Solutions

ENGR 21, Fall 2026.

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Solutions

1 Converting between number systems

1.1 from Decimal to Binary

1. 100

 Answer

0b1100100

2. 1029

 Answer

0b10000000101

3. 476

 Answer

0b111011100

4. 1.4×10^3

 Answer

0b10101111000

1.2 from Binary to Decimal

1. 0b1011011

 Answer

91

2. 0b110010101

 Answer

405

3. 0b1000000001

 Answer

513

4. 0b10111

 Answer

23

1.3 from Decimal to Hexadecimal

1. 1000

 Answer

0x3E8

2. 176

 Answer

0xB0

3. 42678

 Answer

0xA6B6

4. 81

 Answer

0x51

1.4 from Hexadecimal to Decimal

1. 0x14E3

 Answer

5,347

2. 0xA10B

 Answer

41,227

3. 0x1000

 Answer

4,096

4. 0x1010

 Answer

4,112

2 Base systems

2.1 A new base: 7

1. Write down all numbers from 1 to 16 in base 7.

 Solution

The numbers from one to sixteen in base 7 will be: {1, 2, 3, 4, 5, 6, } followed by the number 10, which equals seven, and then 11, which equals eight, and so on. In base 7, the sequence would be

$$\{1, 2, 3, 4, 5, 6, 10, 11, 12, 13, 14, 15, 16, 20, 21, 22\}$$

2. In base 7, what is the meaning of the symbol 66 ? Explain.

 Solution

In base 7, this would need to be interpreted as follows.

$$6 \times 7^0 + 6 \times 7^1 = 6_{10} + 42_{10} = 48_{10}$$

Therefore, the symbol 66 in base 7 corresponds to the number forty-eight.

3. What is the largest possible four-‘digit’ number in base 7? Write the number in words, in decimal form, and in base-7 form.

 Solution

The largest possible four-‘digit’ number in base 7 is 6666_7 . In decimal form, this would be

$$6 \times 7^0 + 6 \times 7^1 + 6 \times 7^2 + 6 \times 7^3 = 6_{10} + 42_{10} + 294_{10} + 2058_{10} = 2400_{10}.$$

In words, it is the number two thousand four hundred.

2.2 Addition of binary and hexadecimal numbers

In grade school, you learned how to add two multi-digit numbers by hand. In case, you've forgotten, [here's](#) a video explaining this to elementary school students.

Your task in this problem is to carry out multi-'digit' addition, by hand, to binary and hexadecimal numbers. In the process, you will see the concept of 'carrying over' a digit (perhaps you learned the term 'regrouping') applies to numbers other than base ten.

As an example, the following sums have been computed for you below.

- $0b110001 + 0b11011$
- $0xAFC + 0x115$

The image shows two handwritten addition problems. The first is in binary: $0b110001 + 0b11011$. The numbers are written in red, with the result 1001100 in blue. The second is in hexadecimal: $0xAFC + 0x115$. The numbers are written in red, with the result $0xC11$ in blue. To the right of each binary addition is a decimal equivalent: $49 + 27 = 76$. To the right of each hexadecimal addition is a decimal equivalent: $2812 + 277 = 3089$.

2.3 Hex

1. $0xE5A + 0x85A$

💡 Answer

5812 in decimal form and $0x16B4$ in hexadecimal form

2. $0xEE + 0x99$

💡 Answer

391 in decimal form and $0x187$ in hexadecimal form

3. $0x100 + 0x20$

💡 Answer

288 in decimal form and $0x120$ in hexadecimal form

4. $0xB0 + 0xB$

 Answer

187 in decimal form and 0xBB in hexadecimal form

2.4 Binary

1. $0b1011011 + 0b11011001$

 Answer

308 in decimal form and 0b100110100 in hexadecimal form

2. $0b10011 + 0b100110001$

 Answer

324 in decimal form and 0b101000100 in hexadecimal form

3. $0b10000000000 + 0b1011111$

 Answer

1119 in decimal form and 0b10001011111 in hexadecimal form

4. $0b11010001001 + 0b110110110$

 Answer

2111 in decimal form and 0b100000111111 in hexadecimal form

3 Conditionals and loops

3.1 Counter variables

Consider the following program, which counts the number of times pin A7 has been tapped and assigns `color1` to pixel number `k`, where `k` is the number of times pin A7 has been tapped. The code crashes if pin A7 is touched more than 10 times.

```
from adafruit_circuitplayground.express import cpx
import time
off = (0,0,0)
color1 = (10,30,10)
color2 = (30,10,10)
cpx.pixels.fill(off)
k = -1
while True:
    time.sleep(0.2)
    if cpx.touch_A7:
        k += 1
        cpx.pixels[k] = color1
        cpx.pixels[k-1] = off
```

Modify this code so that it has the same behavior for up to 10 taps, but for taps 11 through 20, a different color is assigned to the `p`th pixel, where `p` is `k-10`. Your code will still fail after the 21st tap is detected; that's fine.

Solution

```
from adafruit_circuitplayground.express import cpx
import time
off = (0,0,0)
color1 = (10,30,10)
color2 = (30,10,10)
cpx.pixels.fill(off)
k = -1
while True:
    time.sleep(0.2)
    if cpx.touch_A7:
        k += 1
        if k < 10:
            cpx.pixels[k] = color1
            cpx.pixels[k-1] = off
        elif k < 20:
            cpx.pixels[k % 10] = color2
            cpx.pixels[(k % 10)-1] = off
```

3.2 The continue keyword

The following program was written with the intention of printing all numbers except for 9. However, the program currently fails at doing this. What one change will ensure that printing '9' is skipped? Turn in the new code.

```
import time
k = 0
while True:
    time.sleep(0.5)
    k += 1
    print(k)
    if k == 9:
        continue
```

Solution

```
import time
k = 0
while True:
    time.sleep(0.5)
    k += 1
    if k == 9:
        continue
    print(k)
```

3.3 The break keyword

In class, we saw that the following code allows us to terminate the inner `while` loop by touching pin A2.

```
from adafruit_circuitplayground.express import cpx
import time
k = 0
while True:
    time.sleep(1)
    k += 1
    print(f"Outer iteration {k}")
    if k % 3 == 0: # k ÷ 3 has remainder 0
        m = 0
        while True:
            time.sleep(1)
            m += 1
            print(f"Outer iteration {k}, inner iteration {m} ")
            if cpx.touch_A2:
                print("terminating")
                break
```


Change the code above so that it meets the following requirements:

- Touching A2 while in the inner loop should terminate inner loop.
- Touching A2 while in the outer loop should not do anything.
- Touching A7 alone while in the inner loop should now **also** terminate inner loop.
- Touching A7 *while pressing button B* while in the outer loop should terminate the outer loop.

Solution

```
from adafruit_circuitplayground.express import cpx
import time
counter1 = 0
while True:
    time.sleep(1)
    counter1 += 1
    print(f"Outer iteration {counter1}")
    if counter1 % 3 == 0: # k ÷ 3 has remainder 0
        counter2 = 0
        while True:
            time.sleep(1)
            counter2 += 1
            print(f"Outer iteration {counter1}, inner iteration {counter2}")
            if cpx.touch_A2:
                print("terminating inner loop")
                break
            elif cpx.touch_A7 and cpx.button_b:
                print("terminating inner loop")
                break
            elif cpx.touch_A7:
                print("terminating inner loop")
                break
        if cpx.touch_A7 and cpx.button_b:
            print("terminating outer loop")
            break
```

3.4 The time functions

The function `time.monotonic()` can be used to measure intervals of time. For example, the following code measures how long after the red LED comes on the user presses button B.

```
from adafruit_circuitplayground.express import cpx
import time
cpx.red_led = True
start_time = time.monotonic()
while True:
    time.sleep(0.01)
    if cpx.button_b:
```

```
    end_time = time.monotonic()
    interval = end_time - start_time
    break
print(f"Interval was {interval} seconds")
```

Rewrite the code above so that it executes the same functionality without using `break` or `if`.

 Solution

```
from adafruit_circuitplayground.express import cpx
import time
cpx.red_led = True
start_time = time.monotonic()
while not cpx.button_b:
    time.sleep(0.01)
end_time = time.monotonic()
interval = end_time - start_time
print(f"Interval was {interval} seconds")
```

4 The Farmer Was Replaced

For this part of the assignment, you will have to make progress on the game [The Farmer Was Replaced](#). You will be provided a gift card for the game soon.

Start a new game and complete the following tasks in order. Only the marked tasks require a submission; those that are not in bold are not associated with any submission.

Note

It is important that you avoid unlocking any options other than the ones explicitly listed. If you accidentally unlock more features, you should re-start the game.

1. Harvest (grass) five times
2. Unlock **while**
3. Use the **while** loop to harvest 200 units of hay.
4. Describe, in words, what the drone does using each of these snippets of code.

- Option 1

```
while True:
    do_a_flip()
    while True:
        harvest()
```

Answer

Drone does one flip, and after that continues harvesting for ever.

- Option 2

```
while True:
    harvest()
    while True:
        do_a_flip()
```

Answer

Drone harvests once, then does flips for ever.

- Option 3

```
while True:
    harvest()
    do_a_flip()
```

💡 Answer

Drone harvests and flips in turn until the end of time.

- Option 4

```
while True:
    do_a_flip()
    harvest()
```

💡 Answer

Drone harvests and flips in turn until the end of time, but starts with a flip instead of a harvest.

- Option 5

```
while True:
    do_a_flip()
harvest()
```

💡 Answer

Drone does flips for ever, never harvests.

- Option 6

```
while True:
    harvest()
do_a_flip()
```

💡 Answer

Drone harvests for ever, never does a flip.

5. Unlock speed (1/5 only)
6. Write code that executes a flip if a harvest isn't available, and harvests if possible.

💡 Answer

The following code does the trick.

```
if can_harvest():
    harvest()
else:
    do_a_flip()
```

7. Explain why, after unlocking speed, the following code no longer works.

```
while True:
    harvest()
```

💡 Answer

Because there is not enough time for the grass to grow before it is harvested.

8. Unlock bush planting
9. Write a program that causes the drone to keep doing flips as it waits for bushes to grow, and harvests them when they're ready to be harvested. It immediately plants a bush once it is done harvesting.

💡 Answer

The following code does the trick.

```
while True:
    if can_harvest():
        harvest()
        plant(Entities.Bush)
    else:
        do_a_flip()
```

10. Expand the farm. It should now be size 3 x 1.
11. Describe in words what the following code does.

```
while True:
    if can_harvest():
        harvest()
        plant(Entities.Bush)
    else:
        move(North)
```

💡 Answer

This plants bushes on the entire 3 x 1 farm continuously, harvesting them as they grow.

12. We would like to write a program that makes good use of our agricultural resources by planting bushes on two of the plots and grass on the middle plot. It may be helpful to use the following snippet of code for this purpose.

```
clear()
plant(Entities.Bush)
move(North)
move(North)
plant(Entities.Bush)
move(North)
```

Taking inspiration from the above code if necessary, write a script that, when run, continuously plants bushes on the two outer plots, allows grass to grow on the middle spot, and harvests all three plots one by one for ever. The first line in your code should be `clear()`

💡 Solution

The following code does the trick.

```
clear()
while(True):
    if can_harvest():
        # Plant bush and harvest
        harvest()
        plant(Entities.Bush)
        move(North)
        # Harvest grass. grows automatically
        harvest()
        move(North)
        # Plant bush and harvest
        harvest()
        plant(Entities.Bush)
        move(North)
```

13. Unlock carrots.
14. Expand the farm one more time, getting to a 3 x 3 size.
15. Without using `for` loops, achieve the functionality shown by the following gif. It is possible to do so using just 9 lines of code. The first line should be `clear()`.

💡 Solution

The following code does the trick.

```
clear()
while True:
    till()
    move(East)
    till()
    move(East)
    till()
    move(East)
    move(North)
```

16. The following code, when executed, produces the functionality shown in the gif below.

```
clear()
till()
move(East)
till()
move(East)
```

```
till()
move(East)
while True:
    # First row
    if can_harvest():
        harvest()
    plant(Entities.Carrot)
    move(East)
    if can_harvest():
        harvest()
    plant(Entities.Carrot)
    move(East)
    if can_harvest():
        harvest()
    plant(Entities.Carrot)
    move(East)
    move(North)
    # Second row: Grass
    if can_harvest():
        harvest()
    move(East)
    if can_harvest():
        harvest()
    move(East)
    if can_harvest():
        harvest()
    move(East)
    move(North)
    # Third row
    if can_harvest():
        harvest()
    plant(Entities.Bush)
    move(East)
    if can_harvest():
        harvest()
    plant(Entities.Bush)
    move(East)
    if can_harvest():
        harvest()
    plant(Entities.Bush)
    move(East)
    move(North)
```

Modify the above code so that carrots are in the middle row and grass in the bottom row. The bushes should still be in the top row.

 Solution


```
clear()
move(North)
till()
move(East)
till()
move(East)
till()
move(East)
move(South)
while True:
    # First row: Grass
    if can_harvest():
        harvest()
    move(East)
    if can_harvest():
        harvest()
    move(East)
    if can_harvest():
        harvest()
    move(East)
    move(North)
    # Second row: Carrots
    if can_harvest():
        harvest()
    plant(Entities.Carrot)
    move(East)
    if can_harvest():
        harvest()
    plant(Entities.Carrot)
    move(East)
    if can_harvest():
        harvest()
    plant(Entities.Carrot)
    move(East)
    move(North)
    # Third row
    if can_harvest():
        harvest()
    plant(Entities.Bush)
    move(East)
    if can_harvest():
        harvest()
    plant(Entities.Bush)
    move(East)
    if can_harvest():
        harvest()
    plant(Entities.Bush)
    move(East)
    move(North)
```